

PRACTICE PROBLEMS BASED ON PRINCIPAL COMPONENT ANALYSIS-

PROBLEM #1:

PCA Algorithm-

The steps involved in PCA Algorithm are as follows-

Step-01: Get data.

Step-02: Compute the mean vector (μ).

Step-03: Subtract mean from the given data.

Step-04: Calculate the covariance matrix.

Step-05: Calculate the eigen vectors and eigen values of the covariance matrix.

Step-06: Choosing components and forming a feature vector.

Step-07: Deriving the new data set.

Problem-01:

Given data = { 2, 3, 4, 5, 6, 7 ; 1, 5, 3, 6, 7, 8 }.

Compute the principal component using PCA Algorithm.

Solution-

Step-01:

Get data.

The given feature vectors are-

- $x_1 = (2, 1)$
- $x_2 = (3, 5)$
- $x_3 = (4, 3)$
- $x_4 = (5, 6)$
- $x_5 = (6, 7)$
- $x_6 = (7, 8)$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Step-02:

Calculate the mean vector (μ).

Mean vector (μ)

$$= ((2 + 3 + 4 + 5 + 6 + 7) / 6, (1 + 5 + 3 + 6 + 7 + 8) / 6)$$

$$= (4.5, 5)$$

Thus,

$$\text{Mean vector } (\mu) = \begin{bmatrix} 4.5 \\ 5 \end{bmatrix}$$

Step-03:

Subtract mean vector (μ) from the given feature vectors.

- $x_1 - \mu = (2 - 4.5, 1 - 5) = (-2.5, -4)$
- $x_2 - \mu = (3 - 4.5, 5 - 5) = (-1.5, 0)$
- $x_3 - \mu = (4 - 4.5, 3 - 5) = (-0.5, -2)$
- $x_4 - \mu = (5 - 4.5, 6 - 5) = (0.5, 1)$
- $x_5 - \mu = (6 - 4.5, 7 - 5) = (1.5, 2)$
- $x_6 - \mu = (7 - 4.5, 8 - 5) = (2.5, 3)$

Feature vectors (x_i) after subtracting mean vector (μ) are-

-2.5	-1.5	-0.5	0.5	1.5	2.5
-4	0	-2	1	2	3

Step-04:

Calculate the covariance matrix.

The covariance matrix is given by-

$$\text{Covariance Matrix} = \frac{\sum (x_i - \mu)(x_i - \mu)^t}{n}$$

Now,

$$m_1 = (x_1 - \mu)(x_1 - \mu)^t = \begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -2.5 & -4 \end{bmatrix} = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix}$$

$$m_2 = (x_2 - \mu)(x_2 - \mu)^t = \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -1.5 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix}$$

$$m_3 = (x_3 - \mu)(x_3 - \mu)^t = \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} -0.5 & -2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 \\ 1 & 4 \end{bmatrix}$$

$$m_4 = (x_4 - \mu)(x_4 - \mu)^t = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$m_5 = (x_5 - \mu)(x_5 - \mu)^t = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 1.5 & 2 \end{bmatrix} = \begin{bmatrix} 2.25 & 3 \\ 3 & 4 \end{bmatrix}$$

$$m_6 = (x_6 - \mu)(x_6 - \mu)^t = \begin{bmatrix} 2.5 \\ 3 \end{bmatrix} \begin{bmatrix} 2.5 & 3 \end{bmatrix} = \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

Now,

Covariance matrix

$$= (m_1 + m_2 + m_3 + m_4 + m_5 + m_6) / 6$$

On adding the above matrices and dividing by 6, we get-

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$$\text{Covariance Matrix} = \frac{1}{6} \begin{bmatrix} 17.5 & 22 \\ 22 & 34 \end{bmatrix}$$

$$\text{Covariance Matrix} = \begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix}$$

Step-05:

Calculate the eigen values and eigen vectors of the covariance matrix.

λ is an eigen value for a matrix M if it is a solution of the characteristic equation

$$|M - \lambda I| = 0.$$

For an eigenvalue λ ,

$$|M - \lambda I| = 0$$

where

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore,

$$M - \lambda I = \begin{bmatrix} 2.92 - \lambda & 3.67 \\ 3.67 & 5.67 - \lambda \end{bmatrix}$$

2. Take the determinant

For a 2×2 matrix,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$(2.92 - \lambda)(5.67 - \lambda) - (3.67)(3.67) = 0$$

3. Expand the first product

$$\begin{aligned} & (2.92 - \lambda)(5.67 - \lambda) \\ &= 2.92(5.67) - 2.92\lambda - 5.67\lambda + \lambda^2 \\ &= 16.5564 - 8.59\lambda + \lambda^2 \end{aligned}$$

Also,

$$3.67 \times 3.67 = 13.4689$$

Hence,

$$\begin{aligned} 16.5564 - 8.59\lambda + \lambda^2 - 13.4689 &= 0 \\ \lambda^2 - 8.59\lambda + 3.0875 &= 0 \end{aligned}$$

Approximately,

$$\lambda^2 - 8.59\lambda + 3.09 = 0$$

4. Solve the quadratic equation

$$\lambda = \frac{8.59 \pm \sqrt{8.59^2 - 4(1)(3.09)}}{2}$$

This gives approximately:

$$\lambda_1 = 8.22$$

$$\lambda_2 = 0.38$$

The larger eigenvalue, **8.22**, represents the direction of maximum variance and therefore gives the **first principal component**.

Solving this quadratic equation, we get $\lambda = 8.22, 0.38$
Thus, two eigen values are $\lambda_1 = 8.22$ and $\lambda_2 = 0.38$.

Clearly, the second eigen value is very small compared to the first eigen value.
So, the second eigen vector can be left out.

Eigen vector corresponding to the greatest eigen value is the principal component for the given data set. So. we find the eigen vector corresponding to eigen value λ_1 .

We use the following equation to find the eigen vector-
$$MX = \lambda X$$

where-

- M = Covariance Matrix
- X = Eigen vector
- λ = Eigen value

Substituting the values in the above equation, we get-

$$\begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = 8.22 \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

Solving these, we get-

$$2.92X_1 + 3.67X_2 = 8.22X_1$$

$$3.67X_1 + 5.67X_2 = 8.22X_2$$

On simplification, we get-

$$5.3X_1 = 3.67X_2 \quad \dots\dots\dots(1)$$

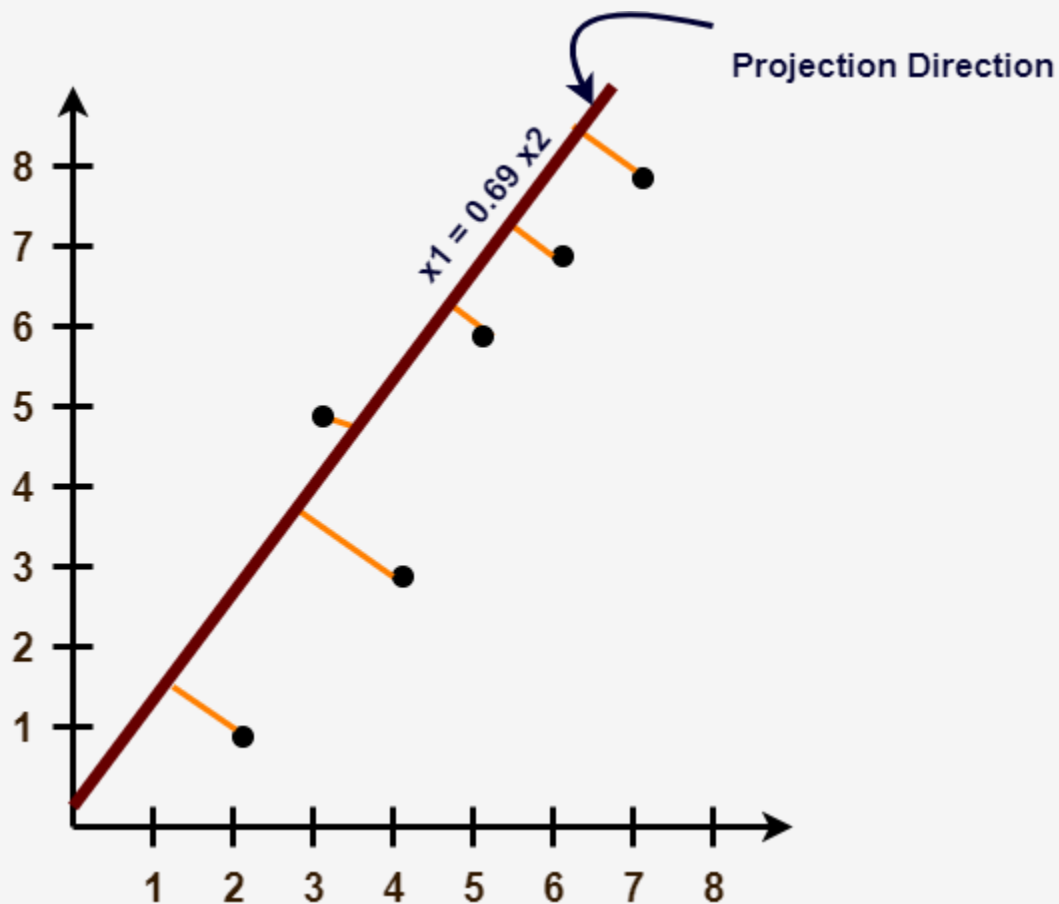
$$3.67X_1 = 2.55X_2 \quad \dots\dots\dots(2)$$

$$X_1 = \frac{2.55}{3.67}X_2$$

$$X_1 \approx 0.69X_2$$

If we choose $X_2 = 1$, then $X_1 = 0.69$. Hence, the eigenvector is proportional to

$$v = \begin{bmatrix} 0.69 \\ 1 \end{bmatrix}$$



PROBLEM #2:

F	⚡ EX1	⚡ EX2	⚡ EX3	⚡ EX4
X1	4	8	13	7
X2	11	4	5	14

Using **population covariance**, divide by $N = 4$.

1. Mean values

$$\bar{X}_1 = 8, \quad \bar{X}_2 = 8.5$$

2. Centered data

$$Z = \begin{bmatrix} -4 & 2.5 \\ 0 & -4.5 \\ 5 & -3.5 \\ -1 & 5.5 \end{bmatrix}$$

3. Population covariance matrix

$$C = \frac{1}{N} Z^T Z$$

$$C = \begin{bmatrix} \frac{42}{4} & \frac{-33}{4} \\ \frac{-33}{4} & \frac{63}{4} \end{bmatrix}$$

Therefore,

$$C = \begin{bmatrix} 10.5 & -8.25 \\ -8.25 & 15.75 \end{bmatrix}$$

4. Find the eigenvalues

$$|C - \lambda I| = 0$$

$$\begin{vmatrix} 10.5 - \lambda & -8.25 \\ -8.25 & 15.75 - \lambda \end{vmatrix} = 0$$

$$(10.5 - \lambda)(15.75 - \lambda) - (-8.25)^2 = 0$$

$$\lambda^2 - 26.25\lambda + 100.125 = 0$$

Solving:

$$\lambda_1 \approx 21.78$$

$$\lambda_2 \approx 4.47$$

The larger eigenvalue is 21.78.

$$a = 1, \quad b = -26.25, \quad c = 97.3125$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{26.25 \pm \sqrt{(-26.25)^2 - 4(1)(97.3125)}}{2}$$

$$\lambda = \frac{26.25 \pm \sqrt{299.8125}}{2}$$

$$\sqrt{299.8125} \approx 17.315$$

$$\lambda_1 = \frac{26.25 + 17.315}{2}$$

$$\lambda_1 \approx 21.78$$

$$\lambda_2 = \frac{26.25 - 17.315}{2}$$

$$\lambda_2 \approx 4.47$$

$$\lambda_1 = 21.78, \quad \lambda_2 = 4.47$$

5. Find the eigenvector for $\lambda_1 = 21.78$

$$(C - \lambda I)v = 0$$

Using the first equation:

$$(10.5 - 21.78)X_1 - 8.25X_2 = 0$$

$$-11.28X_1 - 8.25X_2 = 0$$

$$X_2 = -1.367X_1$$

Choose $X_1 = 1$:

$$v = \begin{bmatrix} 1 \\ -1.367 \end{bmatrix}$$

Normalize it:

$$\|v\| = \sqrt{1^2 + (-1.367)^2} \approx 1.694$$

$$v_1 = \frac{1}{1.694} \begin{bmatrix} 1 \\ -1.367 \end{bmatrix}$$

Hence,

$$v_1 \approx \begin{bmatrix} 0.590 \\ -0.807 \end{bmatrix}$$